

Now that we have seen what the methodology and interpretation for RMANOVA . Let's look at the methodology for linear Mixed Model (LMM), using the pilot dataset.

Obs	Patient_ID	Group	Age	Week_0	Week_4	Week_8	Week_12
1	P001	Treatment	45	82.5	78.1	70.4	62.1
2	P002	Treatment	52	79.0	74.3	68.0	60.5
3	P003	Control	38	85.2	84.0	86.1	84.9
4	P004	Treatment	61	91.4	85.0	.	71.3
5	P005	Control	49	75.0	76.2	74.8	77.0
6	P006	Treatment	34	88.1	80.2	73.5	65.4
7	P007	Control	55	80.3	81.5	79.9	.
8	P008	Control	42	86.0	85.5	87.2	86.5
9	P009	Treatment	47	93.7	88.4	81.0	72.8
10	P010	Control	50	78.5	.	79.0	80.2

For LMM we can see the patients 4, 7 and 10 have missing data for events; week 8, week 12 and week 4. Based on the methodology of LMM, these 3 patients will be retained but the exact events missed will not be counted. So instead of 40 data points, LMM will focus on the 37 data points available for analyses.

The first step is to determine which covariance structure provides the best fit for the data by comparing First-Order Autoregressive (AR1), Compound Symmetry (CS), and Unstructured (UN) structures.

```
proc mixed data = clinical_long method = ml;
  class Patient_ID Group Age Week (ref = first);
  model Outcome = Group Week Group*Week / s;
  repeated Week / subject = Patient_ID type = cs;
  title "Model Fit: Compound Symmetry (CS)";
run;
title; /* Clear titles */
```

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
CS	Patient_ID	24.9434
Residual		0.7383

Fit Statistics	
-2 Res Log Likelihood	124.2
AIC (Smaller is Better)	128.2
AICC (Smaller is Better)	128.6
BIC (Smaller is Better)	128.8

Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > Chi Sq
1	66.14	<.0001

```
proc mixed data = clinical_long method = ml;
  class Patient_ID Group Age Week (ref = first);
  model Outcome = Group Week Group*Week / s;
  repeated Week / subject = Patient_ID type = ar(1);
  title "Model Fit: Autoregressive (AR1)";
run;
title; /* Clear titles */
```

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
AR(1)	Patient_ID	0.9757
Residual		25.7635

Fit Statistics	
-2 Res Log Likelihood	126.3
AIC (Smaller Is Better)	130.3
AICC (Smaller Is Better)	130.7
BIC (Smaller Is Better)	130.9

Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > Chi Sq
1	64.02	<.0001

```
proc mixed data = clinical_long method = ml;
class Patient_ID Group Age Week (ref = first);
model Outcome = Group Week Group*Week / s;
repeated Week / subject = Patient_ID type = un;
title "Model Fit: Unstructure (UN)";
run;
title; /* Clear titles */
```

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
UN(1,1)	Patient_ID	22.1135
UN(2,1)	Patient_ID	21.8902
UN(2,2)	Patient_ID	22.4234
UN(3,1)	Patient_ID	25.5499
UN(3,2)	Patient_ID	25.2085
UN(3,3)	Patient_ID	30.2895
UN(4,1)	Patient_ID	24.9428
UN(4,2)	Patient_ID	25.0432
UN(4,3)	Patient_ID	29.1587
UN(4,4)	Patient_ID	29.3840

Fit Statistics	
-2 Res Log Likelihood	116.1
AIC (Smaller Is Better)	136.1
AICC (Smaller Is Better)	148.4
BIC (Smaller Is Better)	139.2

Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > Chi Sq
9	74.17	<.0001

Assessing the fit statistics for all the covariances run above: the AIC for the 128.2 (CS, 130.3 (AR1), and 136.1 (UN). The result indicates the compound symmetry fit is the best model

because it mimics the reality of the data efficiently. Maximum Likelihood (ML) estimation is used when comparing fixed effects via Likelihood Ratio Tests, whereas Restricted Maximum Likelihood (REML) is preferred for unbiased estimation of variance components in the final model.

Having the best fit, the next step is to assess if the likelihood that the model is better accounts for all effects or just the basic influences or variables.

```
proc mixed data=clinical_long method=ml;
  class Patient_ID Group Age Week (ref=first);
  model Outcome = Group Week / s;
  repeated Week / subject=Patient_ID type=cs;
  title "Null/Reduced statistics";
run;
title; /* Clear titles */
```



Fit Statistics	
-2 Log Likelihood	232.3
AIC (Smaller is Better)	246.3
AICC (Smaller is Better)	250.2
BIC (Smaller is Better)	248.4

```
proc mixed data=clinical_long method=ml;
  class Patient_ID Group Age Week (ref=first);
  model Outcome = Group Week Group*Week / s;
  repeated Week / subject=Patient_ID type=cs;
  title "Full Model statistics";
run;
title; /* Clear titles */
```



Fit Statistics	
-2 Log Likelihood	133.0
AIC (Smaller is Better)	153.0
AICC (Smaller is Better)	161.5
BIC (Smaller is Better)	156.1

Comparing the Likelihood Ratio Test (LRT):

Test Statistic (χ^2):

$$\Delta(-2LL) = (-2LL_{\text{Null}}) - (-2LL_{\text{Full}})$$

$$= 232.3 - 133.0$$

$$= 99.3, df=3$$

$$\chi^2(3) = 99.3, p < 0.0001,$$

Looking at the 2 Log likelihood the full model statistics give the model more robust output and fit.

Let's look at the chosen model to access the fit statistics under the REML

```
proc mixed data = clinical_long method = reml;
  class Patient_ID Group Age Week (ref = first);
  model Outcome = Group Week Group*Week / s solution cl;
  repeated Week / subject = Patient_ID type =cs r rcorr;
  title "Final Model - REML Estimation";
run;
```

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
CS	Patient_ID	24.9434
Residual		0.7383

Fit Statistics	
-2 Res Log Likelihood	124.2
AIC (Smaller is Better)	128.2
AICC (Smaller is Better)	128.8
BIC (Smaller is Better)	128.8

Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
1	66.14	<.0001

The fit for this analysis is smaller at REML model statistic =128.2 compared to 153.0 for ML, hence better fit.

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
Group	1	8	1.75	0.2228
Week	3	21	219.81	<.0001
Group*Week	3	21	264.10	<.0001

The F-value is the snap shot of data detailing the effect between the dependent variables and the outcome. The effect of time and time interaction with treatment groups is significant at an (F-value=219.81, $p<0.001$) and (F-value=264.10, $p<0.001$) respectively. The treatment effect is not significant and does not affect the outcome F-value= 1.75, $p=0.2228$)

Solution for Fixed Effects										
Effect	Group	Week	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper
Intercept			86.9400	2.2664	8	38.36	<.0001	0.05	81.7138	92.1662
Group	Control		-5.9400	3.2051	8	-1.85	0.1010	0.05	-13.3310	1.4510
Group	Treatment		0	-	-	-	-	-	-	-
Week		Week_12	-20.5200	0.5434	21	-37.76	<.0001	0.05	-21.6501	-19.3899
Week		Week_4	-5.7400	0.5434	21	-10.56	<.0001	0.05	-6.8701	-4.6099
Week		Week_8	-12.6307	0.5869	21	-21.52	<.0001	0.05	-13.8512	-11.4102
Week		Week_0	0	-	-	-	-	-	-	-
Group*Week	Control	Week_12	21.5093	0.8001	21	26.88	<.0001	0.05	19.8454	23.1731
Group*Week	Control	Week_4	5.9880	0.8001	21	7.48	<.0001	0.05	4.3242	7.6518
Group*Week	Control	Week_8	13.0307	0.7998	21	16.29	<.0001	0.05	11.3673	14.6941
Group*Week	Control	Week_0	0	-	-	-	-	-	-	-
Group*Week	Treatment	Week_12	0	-	-	-	-	-	-	-
Group*Week	Treatment	Week_4	0	-	-	-	-	-	-	-
Group*Week	Treatment	Week_8	0	-	-	-	-	-	-	-
Group*Week	Treatment	Week_0	0	-	-	-	-	-	-	-

The Intercept is ($\beta=86.94$, $p<0.001$). This value represents your exact clinical starting line: the predicted average outcome score for a patient in the Treatment group at Week 0. Let's look at how the model tracks the deviations moving forward from baseline:

The Treatment Group's Main Trajectory (Reference Rows)

Because **Treatment** and **Week0** are the internal reference values (set to 0), we can read the main **Week** estimates to see exactly how the treated patients progressed:

- Week 4 ($\beta=-5.7400$, $p<0.001$): By Week 4, the treatment group dropped an average of 5.74 points from baseline ($p < .0001$), resulting in a predicted mean score of 81.20.
- Week 8 ($\beta=-12.6828$, $p<0.001$): By Week 8, they dropped 12.68 points below baseline ($p < .0001$), resulting in a predicted mean score of 74.26.
- Week 12 ($\beta=-20.5200$, $p<0.001$): By Week 12, the treated patients plummeted a massive 20.52 points lower than their starting scores ($p < .0001$), resulting in a predicted mean score of 66.42.
- *Main Trajectory Conclusion:* The treatment group experienced a powerful, highly significant, and linear therapeutic reduction across the entire 12-week timeline.

The Control Group's Baseline Separation

- **Group Control** ($\beta=-5.9400$, $p = 0.1014$): This is highly revealing. It shows that at Week 0, the Control group actually started out averaging 5.94 points lower than the Treatment group. However, with a p-value of 0.1014, this initial baseline difference is *not* statistically significant. This indicates that the baseline randomization held up fine.

The **Group*Week** Interaction (The Difference of Differences)

These parameters show how much the Control group's path diverged from the treatment group's downward trajectory over time:

- **Control * Week 4** ($\beta=5.9888$, $p<0.001$): While the treatment group exhibited a significant score reduction by Week 4, the control group remained relatively stable. This positive interaction estimate ($\beta = 5.99$) indicates that by Week 4, the control group scored 5.99 points higher relative to baseline than the treatment group. .
- **Control * Week 8** ($\beta=13.0828$, $p<0.001$): By Week 8, the group trajectories diverged further, with the control group demonstrating a 13.08-point higher relative score compared to the treatment group's decline.
- **Control * Week 12** ($\beta=21.7313$, $p<0.001$): By the final assessment, the interaction effect reached its peak. Relative to their respective baseline scores, the control group finished 21.73 points higher than the treatment group ($p < .0001$), confirming a statistically significant widening gap over time.